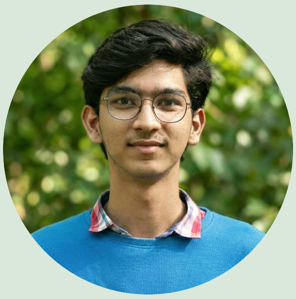

LAKSHMI OMKARESWAR THUMMAGUNTA

Electronics and Communication Engineering, VLSI & Embedded Systems Enthusiast

omkar@yahoo.com | +91 - 90002 45274 | LinkedIn.com/omkar-tl | Nellore, AP, India

SUMMARY

Dedicated engineering student with practical experience in innovative projects, including a Mini RISC Processor, a secure Fingerprint-Based Vehicle Starting System, and an AI-Powered OLED Display Interface. Proficient in Verilog HDL, embedded systems design, and microcontroller programming, eager to leverage technical skills in real-world applications.

PROJECTS

AI-Based Automatic Bike Cleaning System
11/2025 – 03/2026
- Designed FSM-based automated control system using sensor inputs and microcontroller.
- Implemented real-time hardware interfacing (motors, pumps, sensors) via GPIO.
- Optimized system efficiency (70% water saving, 2–3 min cycle).
- Finalist – **InnoWAH (PALS)**.

Mini RISC PROCESSOR
09/2025 – 10/2025
- Developed a Single-Cycle 8-bit RISC Processor using Verilog HDL, which includes a comprehensive ALU, register file, control unit, and memory interface, demonstrating the ability to execute a variety of operations within a single clock cycle.

AI-Powered OLED Display Interface ESP32
04/2025 – 04/2025
- Created a cutting-edge embedded system that combines OpenAI API with a 128x64 OLED display, employing efficient 12C communication along with optimized memory and power management for enhanced edge-AI applications.

TECHNICAL SKILLS

Verilog HDL | Embedded C | Python | C - Language | C ++ | VLSI & ASIC Design | FPGA System Design (Vivado) | IoT Development | Wireless Communication | Microcontroller Programming (8051/ESP32) | P-SPICE | LT-SPICE Simulation

LANGUAGES KNOWN

Telugu • English • Hindi • Tamil • Japanese

PROFESSIONAL EXPERIENCE

Andhra Pradesh Power Generation Corporation Limited
Internship Trainee
11/2023 – 05/2024 | Nellore, Andhra Pradesh, India
- Gained hands-on experience in power plant operations, electrical systems, and thermal energy processes.
- Contributed to efficiency analysis, maintenance support, and data collection, developing a strong understanding of power generation systems.
- Collaborated with industry professionals to strengthen problem-solving abilities and gain practical exposure to power systems, energy management, and technical analysis.

EDUCATION

B.E in Electronics and Communication Engineering
Sri Chandrasekharendra Saraswathi Viswa Mahavidyalaya
2023 – Present
- CGPA - 8.60

Diploma in Electronics and Communication Engineering
Audisankara College of Engineering & Technology
2021 – 2024
- Percentage - 86%

CERTIFICATES

VLSI Design – Internshala
- Completed an 8-week VLSI Design training covering HDL fundamentals, combinational & sequential logic, FSMs, and FPGA-based system design

ACHIEVEMENTS

Finalist – InnoWAH Challenge (PALS)
PALS - IIT Alumni initiative
28/03/2026
- **Selected as Finalist – InnoWAH (PALS)**, for designing a smart, sensor-driven automated system with real-world impact.

Special Prize for Innovative Engineering Solution - Think2Impact
PALS - IIT Alumni initiative
14/03/2026
- Successfully presented and validated the project in a **competitive, industry-oriented evaluation round**.